

WardLink PQ

Lightweight, quantum-safe monitoring for India’s municipal water tanks

Project report for supervisor review

Domain	Security in Futuristic Technologies
Problem statement	Lightweight Post-Quantum Messaging (DSCI)
Prepared by	CB
Date	5 October 2026
Status	Working software prototype with live dashboard; 40 of 40 field scenarios passing

The project in one line

Sensors on ward water tanks send sealed, quantum-safe readings to the section office, which checks each reading against the Indian drinking-water standard (IS 10500) and the tank’s physics, and tells the field crew exactly what to do.

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1. Executive summary

WardLink PQ is a monitoring link between a sensor on a ward water tank and the section office. Readings travel through a channel that stays confidential and authentic even against future quantum computers, yet it is light enough for low-cost, battery-powered hardware and narrow long-range radios. At the office, each reading is checked against the Indian drinking-water standard (IS 10500:2012) and against what the tank can physically do, and turned into a clear work order: send a tanker, hold the safe-water notice and take a sample, walk the outlet main for a leak, or inspect a faulty sensor.

The project answers the DSCI problem statement “Lightweight Post-Quantum Messaging” with a concrete, India-specific use: securing the telemetry of intermittent piped-water supply.

Key results, measured on the prototype

- Quantum-safe connection set-up (handshake) of **4,544 bytes**: 49% smaller than the usual signature-based post-quantum design (8,978 bytes) and 78% smaller than fully post-quantum TLS 1.3 (20,767 bytes), with quantum-safe secrecy and identity.
- Each reading travels as a **40-byte sealed capsule** with its own key: a single LoRa radio frame, and a stolen device cannot decrypt anything it sent earlier.
- **40 of 40 real-life scenarios pass**: NIST conformance, power cuts, stolen and cloned boards, faulty sensors and 20 tanks at once. 2,400 corrupted messages caused no crash, and no reading was lost through outages or restarts.
- Two real reliability bugs were found by deliberately attacking the gateway, and fixed.

Status and next step. The complete software prototype runs on a laptop with a live, animated dashboard. The next step is to port the sensor firmware to ESP32-S3 hardware and pilot the system on one ward tank with the municipal water department.

2. Background and problem

2.1 The DSCI problem statement

The Data Security Council of India (DSCI) problem statement “Lightweight Post-Quantum Messaging” (domain: Security in Futuristic Technologies) asks for messaging that remains confidential and forward-secret in the quantum era and is light enough for mobile and IoT devices. Its deliverables are a prototype using NIST post-quantum algorithms (Kyber, now standardised as ML-KEM, and Dilithium, now ML-DSA), a key exchange optimised for constrained devices, benchmarks against classical TLS, and a demonstration on constrained devices.

2.2 Why quantum safety matters now

Today’s secure connections agree on keys using elliptic-curve cryptography, which a sufficiently large quantum computer running Shor’s algorithm could break. Traffic recorded today can be stored and decrypted later, an attack known as “harvest now, decrypt later”. In August 2024 NIST published the first post-quantum standards: FIPS 203 (ML-KEM) for key agreement and FIPS 204 (ML-DSA) for signatures. Field devices installed now may stay in service for many years, often without anyone visiting to update their keys, so they need protection that will still hold when quantum computers arrive.

2.3 The ground reality in India's water supply

Many Indian towns and villages receive piped water for only a few hours a day. It is stored in overhead tanks and drawn down by households in between. Tank levels and quality alerts often travel by phone call, SMS, WhatsApp groups or open internet dashboards. Such messages can be faked ("tank full, water safe"), read by anyone, and are usually trusted without checking whether the number is even physically possible. AI tools have also made it cheaper to find and exploit exposed IoT devices.

2.4 Our problem statement

Build a low-cost monitoring link for ward water tanks that:

- keeps readings **secret and authentic** against today's attackers and future quantum computers;
- runs on **battery-powered hardware** and narrow long-range radios;
- **never loses a reading** during outages, restarts or attacks;
- turns readings into **correct, explainable actions** for the field crew.

3. What WardLink PQ is

The system has three parts:

- **Sensor unit (tank roof).** An ESP32-S3 microcontroller with an ultrasonic level sensor, turbidity, TDS (total dissolved solids) and pH probes, temperature sensors, a solar-charged battery and a radio (Wi-Fi in the prototype; LoRa or NB-IoT in the field).
- **Gateway (section office).** Verifies, stores and evaluates every reading. A laptop today; a Raspberry Pi or office PC in the field.
- **Engineer dashboard.** A web page that also opens on a phone. It shows a live, animated cutaway of the system, the current work order, and tools to demonstrate attacks and run test scenarios.

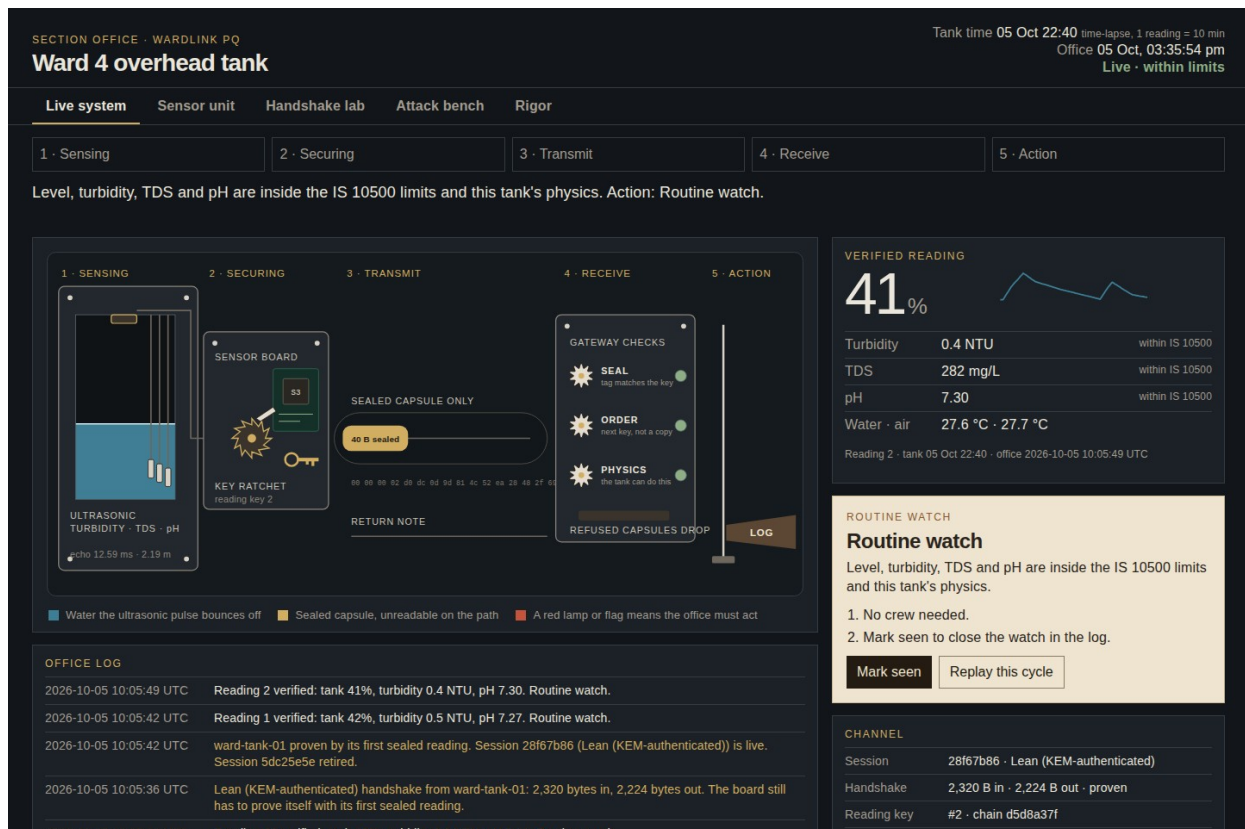


Figure 1. Live system view. Each verified reading animates through five stations: sensing, securing, transmission, the gateway's three checks, and the action flag. The right side shows the reading, the work order and the channel status.

4. Applications

Application	What WardLink PQ provides
Ward overhead tanks and elevated service reservoirs	Live level, refill tracking, and a tanker forecast before the tank runs dry.
Drinking-water quality surveillance	Turbidity, TDS and pH checked against IS 10500, with “hold the safe-water notice” and sampling instructions.
Leak and theft detection	A night-flow test flags leaks and unauthorised connections on the outlet side.
Rural piped-water schemes (for example, villages under the Jal Jeevan Mission)	The same unit monitors village tanks over long-range radio on solar power.
Dams and reservoirs (read-only)	Trusted level telemetry; gate and pump control stay on separate systems.
Other critical telemetry	The same secure channel suits any small sensor in smart-city, industrial or agricultural settings.

5. Advantages

5.1 Security

- **Quantum-safe secrecy and identity.** ML-KEM-768 (FIPS 203) is combined with classical X25519, so an attacker must break both.
- **A new key for every reading.** A stolen board cannot decrypt anything it sent before the theft.
- **Defence in depth.** Cryptography proves who sent a reading; the physics and IS 10500 checks decide whether to believe it and how to act.
- **Stolen-device handling.** Clone detection, revocation and on-site re-enrolment.

5.2 Lightweight

- **4,544-byte handshake**, half the signature-based design. No certificates or PKI are needed, because devices are enrolled in person.
- **20-byte reading, 40 bytes sealed**: one LoRa frame, against four frames for the same reading as text.
- **About 1.3 ms** of computation for both sides of a handshake on a laptop, and no signature code on the sensor.

5.3 Reliability

- **No lost readings**. Each reading stays on the board until the office confirms it.
- **Survives** power cuts, dead links, gateway restarts and connection floods (each one tested).
- **A watchdog** flags a tank that goes silent.

5.4 Practical and verified

- **Low-cost, commonly available components**, solar powered.
- **Actions follow the Indian standard** and the realities of intermittent supply.
- **An explainable dashboard** that engineers and supervisors can follow without a cryptography background.
- **Verified**: 40 field scenarios and 35 unit tests, all passing; official NIST test vectors; 2,400 fuzzed messages.

6. How it works

6.1 One reading, end to end

Stage	What happens	Key detail
1. Sensing	The ultrasonic sensor times its echo from the water surface; the probes read turbidity, TDS, pH and temperature.	Distance = speed of sound × echo time ÷ 2, corrected for air temperature.
2. Securing	The reading is packed into 20 bytes and sealed with ChaCha20-Poly1305 under a key used only once.	40-byte sealed capsule.
3. Transmission	The capsule waits in the board's outbox and is sent over the radio.	It stays until the office's receipt arrives.
4. Receiving	The gateway checks the seal, that the key is the next in line (not a replay), and that the tank could physically do this.	Stored once in the database; a sealed receipt goes back.
5. Action	Rules turn the reading into a work order; the engineer marks it seen.	A sealed "seen" note returns to the board.

6.2 The sensor unit

The sensor unit sits on the tank roof in an IP65 enclosure. The ultrasonic transducer is fixed under the roof and never touches the water; the water-quality probes hang into the tank. Every ten minutes in the field (faster in the time-lapsed demo) the board runs one sampling cycle: wake, measure air temperature, send an ultrasonic ping, read turbidity, TDS, pH and water temperature, pack the 20-byte record, seal it, send it over the radio, and sleep.

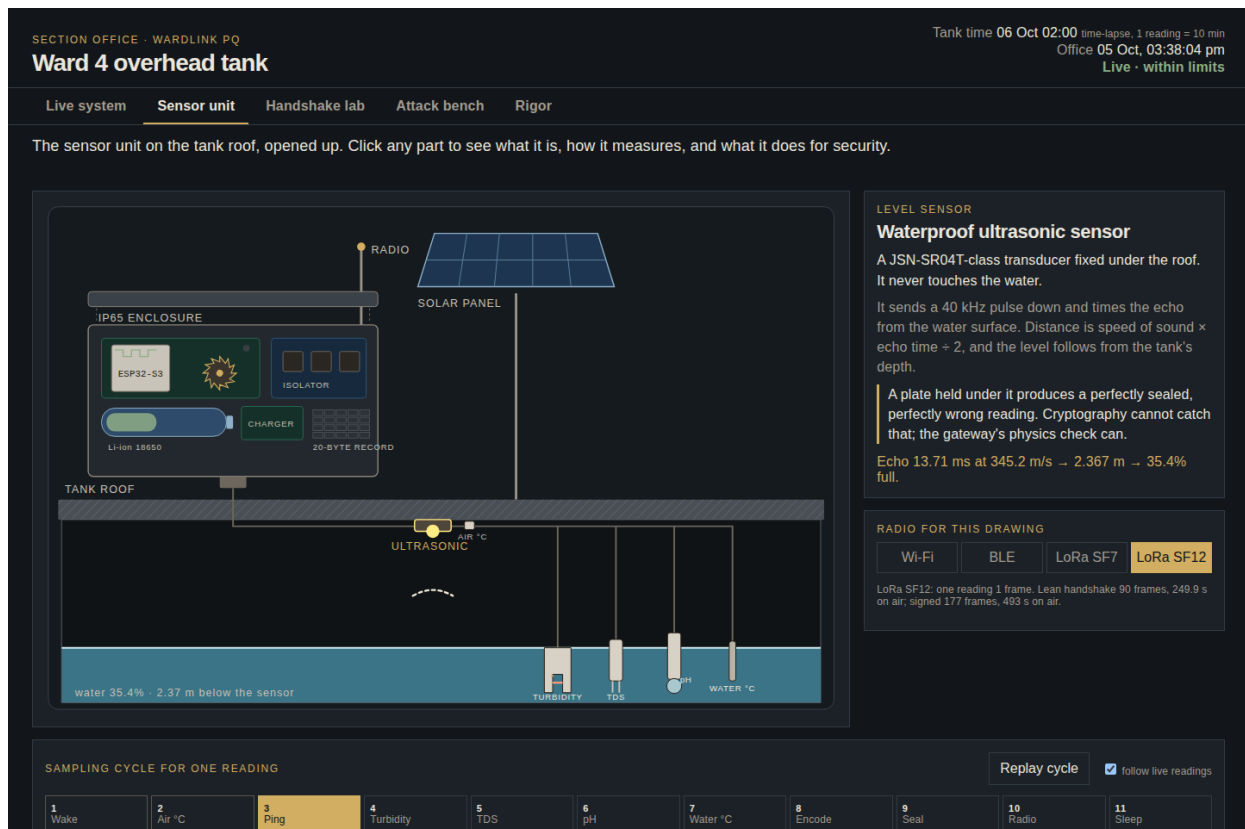


Figure 2. Sensor unit view: an exploded drawing of the roof unit. Every part is clickable and explains what it measures and what it does for security; the sampling cycle below animates with live values.

6.3 The lean quantum-safe handshake

Before sending readings, the board and the gateway agree on a fresh session key. Because the board was enrolled in person at the ward office, each side already holds the other's long-term public keys. A conventional design would have each side sign the handshake with ML-DSA-65, at 3,309 bytes per signature, which is three-quarters of all handshake bytes. WardLink PQ instead lets each side prove its identity by opening ML-KEM capsules addressed to its enrolled key, the idea behind KEMTLS and Post-Quantum Noise in the research literature. Six shared secrets, from ML-KEM-768 and X25519, are mixed into the session key, and the temporary keys are deleted afterwards.

Message	Sent by	Contents	Size
Hello	Sensor	Device id; temporary X25519 and ML-KEM keys; a capsule for the gateway's enrolled key	2,320 B
Welcome	Gateway	Temporary X25519 key; a capsule for the sensor's temporary key; a capsule for its enrolled key; a confirmation tag	2,224 B

- **The gateway proves itself:** its confirmation tag can only be computed with the gateway's enrolled keys.
- **The sensor proves itself:** its first sealed reading can only be produced with the sensor's enrolled keys.
- **Hybrid:** reading or impersonating needs both ML-KEM and X25519 broken.
- **Forward secrecy:** the temporary keys are deleted, so a later key theft cannot open recorded sessions.

6.4 A new key for every reading

After the handshake, each reading is sealed with its own key, taken from a one-way chain. Once used, a key is deleted and cannot be recomputed from later keys. A thief who steals the board and reads its memory can therefore read nothing sent earlier, and a fresh handshake (every 36 readings, about six hours, or on demand) locks the thief out of later readings too.

6.5 No lost readings

Every reading goes into an outbox on the board and leaves it only when the gateway's sealed receipt arrives. During an outage the board keeps measuring and marks readings as buffered, keeping their original time; on reconnection it sends them in order. If a receipt is lost, the board resends and the gateway recognises the copy by its time and stores it once. All readings and events go into a database, so a gateway restart loses neither data nor the context the rules need.

6.6 Decision rules

Check	Rule	Action
Turbidity	IS 10500: 1 NTU acceptable, 5 NTU permissible	Above 5 NTU: hold the safe-water notice and take a sample. A single spike waits one reading for confirmation unless it is above 15 NTU.
pH	6.5 to 8.5, no relaxation	Outside the range: lab sample before the next supply.
TDS	500 mg/L acceptable, 2,000 mg/L permissible	Above 2,000 mg/L: reject the supply for drinking.
Tank physics	The level cannot exceed the overflow pipe, rise faster than the inlet, rise with the main closed, or fall faster than any draw	Treat the level as untrusted and inspect the sensor.
Leak	Night draw between 01:00 and 04:00 above 2.4% per hour (homes use under 1%)	Walk the outlet main; look for unauthorised taps.
Tanker	Projected level below 10% before the next supply window	Book a tanker before the predicted time.
Sensor faults	No echo, object in the blind zone, frozen reading, probes out of the water, impossible values	Name the fault; do not act on that reading.

Water-quality limits are taken from IS 10500:2012, Table 1.

6.7 Device lifecycle

- **Enrolment in person:** keys are created at the ward office; the gateway roster stores public keys only.
- **Periodic re-key:** a fresh handshake every 36 readings, or on demand.
- **Clone alarm:** two boards with the same keys are detected by their overlapping sessions; the tank is quarantined and the overlapping readings are marked suspect.
- **Revoke and re-enrol on site:** new keys are issued; stolen keys stop working.
- **Watchdog and rate limit:** a silent tank is flagged; at most six handshakes per minute per tank.

7. Security analysis

Threat	Example	Defence	Verified by
Future quantum decryption	Traffic recorded today is decrypted later	ML-KEM-768 + X25519 hybrid handshake	Handshake and NIST vector scenarios
Tampering	One bit changed in transit	Poly1305 seal on every capsule	Bit-flip scenario
Replay	An old "tank full" sent again	One-time keys in strict order	Replay scenario

Threat	Example	Defence	Verified by
Impostor device	A board with its own keys claims the tank's id	Enrolled keys	Impostor scenario
Unknown device	A board that was never enrolled	Gateway roster	Unlisted-board scenario
Fake gateway	An attacker pretends to be the office	Gateway's confirmation tag	Fake-gateway scenario
Stolen device	Memory dumped to read old traffic	Per-reading keys; re-key; re-enrolment	Stolen-board scenario
Cloned device	A second board built from stolen keys	Clone alarm, quarantine, re-enrolment	Clone and re-enrol scenarios
Physical spoofing	A plate held under the level sensor	Tank-physics check	Plate-under-sensor scenario
Faulty instruments	Frozen sensor, no echo, bubble on a probe	Fault rules and confirmation	Sensor-fault scenarios
Network abuse	Stalled or flooding connections	Threads, timeouts, connection cap, rate limit	Transport-fault scenarios

SECTION OFFICE · WARDLINK PQ

Ward 4 overhead tank

Tank time 05 Oct 22:40

time-lapse, 1 reading = 10 min

Office 05 Oct, 03:35:56 pm

Live · within limits

Live system

Sensor unit

Handshake lab

Attack bench

Rigor

Try to break it. Every attack runs against the live gateway and real ciphertext; the result comes back from the backend, not from a script.

ON THE WIRE

Flip one bit in transit
Changes a single bit of the next capsule somewhere between the tank and the office.
STOPPED BY THE SEAL

Flip a bit in the next capsule

Replay a recorded capsule
Records a genuine capsule and sends the copy again, hoping the office logs an old reading as new.
STOPPED BY THE RATCHET ORDER

Replay the last capsule

Impostor board
Builds its own board, names it ward-tank-01, and sends "tank 97%, water clean".
STOPPED BY THE ENROLLED KEYS

Send the impostor

Unlisted board
Connects a board that was never enrolled at the ward office.
STOPPED BY THE ROSTER

Connect tank-77

Steal the board
Unbolts the unit from the roof and dumps its memory to read traffic it recorded earlier.
LIMITED BY THE RATCHET, HEALED BY RE-KEY

Steal the board now

Re-key now

Clone the board
Builds a second board from keys dumped out of a stolen unit. Its capsules are genuine, so cryptography alone cannot tell the two apart.
CAUGHT BY THE CLONE ALARM, FIXED BY RE-ENROLLING

Connect the clone

Revoke

Re-enroll on site

Radio outage
Not an attacker: rain fade, a dead repeater or a power cut on the roof takes the link away for a while.

IN THE TANK

Night leak
A joint on the outlet main starts leaking (or someone taps the line).
CAUGHT BY THE TANKER FORECAST AND THE NIGHT-FLOW TEST

Start the leak

Dirty supply
Muddy water enters through the inlet after a pipe repair upstream.
CAUGHT BY THE IS 10500 LIMITS

Contaminate the supply

Plate under the sensor
Holds a plate under the ultrasonic sensor so the tank looks full. Every capsule stays genuine and correctly sealed.
CAUGHT ONLY BY THE PHYSICS CHECK

Block the sensor

Frozen level sensor
The sensor's firmware hangs and keeps repeating its last distance.
CAUGHT BY THE STUCK-SENSOR RULE AFTER SIX READINGS

Freeze the sensor

No echo
Condensation covers the transducer, so its pulse never comes back.
CAUGHT AS A SENSOR FAULT; NO TANKER IS SENT ON IT

Fog the transducer

Bubble on the turbidity probe
One reading jumps above 5 NTU because a bubble crossed the optics.
WAITS FOR A SECOND READING INSTEAD OF A FALSE ALARM

Send a bubble

Back to normal
Fix the leak, remove the plate, clean and restart the sensors, clean supply.
RESETS THE TANK

Figure 3. Attack bench. Each button runs a real attack or fault against the live gateway and shows the gateway's response; the table at the bottom maps every attack to the layer that stops it.

8. Results

8.1 Handshake benchmark

All five handshakes were measured on the same laptop with mutual authentication: TLS through OpenSSL 3.5.5, and WardLink through the project's own code.

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Handshake	Secrecy	Identity	Bytes	Time	SF12 frames	SF12 airtime
TLS 1.3 classical	classical	classical	1,685	≈1.1 ms	34	93 s
TLS 1.3 hybrid	quantum-safe	classical	4,010	≈2.3 ms	80	221 s
WardLink lean	quantum-safe	quantum-safe	4,544	≈1.3 ms	90	250 s
WardLink signed	quantum-safe	quantum-safe	8,978	≈3.3 ms	177	493 s
TLS 1.3 post-quantum	quantum-safe	quantum-safe	20,767	≈5.0 ms	408	1,139 s

Bytes are exact. Times are medians for both sides together and vary between runs. Radio cost uses LoRaWAN IN865 at its slowest setting (SF12, 51-byte frames) and the Semtech time-on-air formula. TLS uses one self-signed certificate per side; a real certificate chain would add more.

The lean handshake gives quantum-safe identity for 13% more bytes than today's hybrid TLS, which still uses classical identity, and at less than a quarter of the size of fully post-quantum TLS. On LoRa at its slowest setting a handshake still takes about four minutes of airtime, so the board re-keys rarely and uses the per-reading key chain in between; each reading is a single frame.

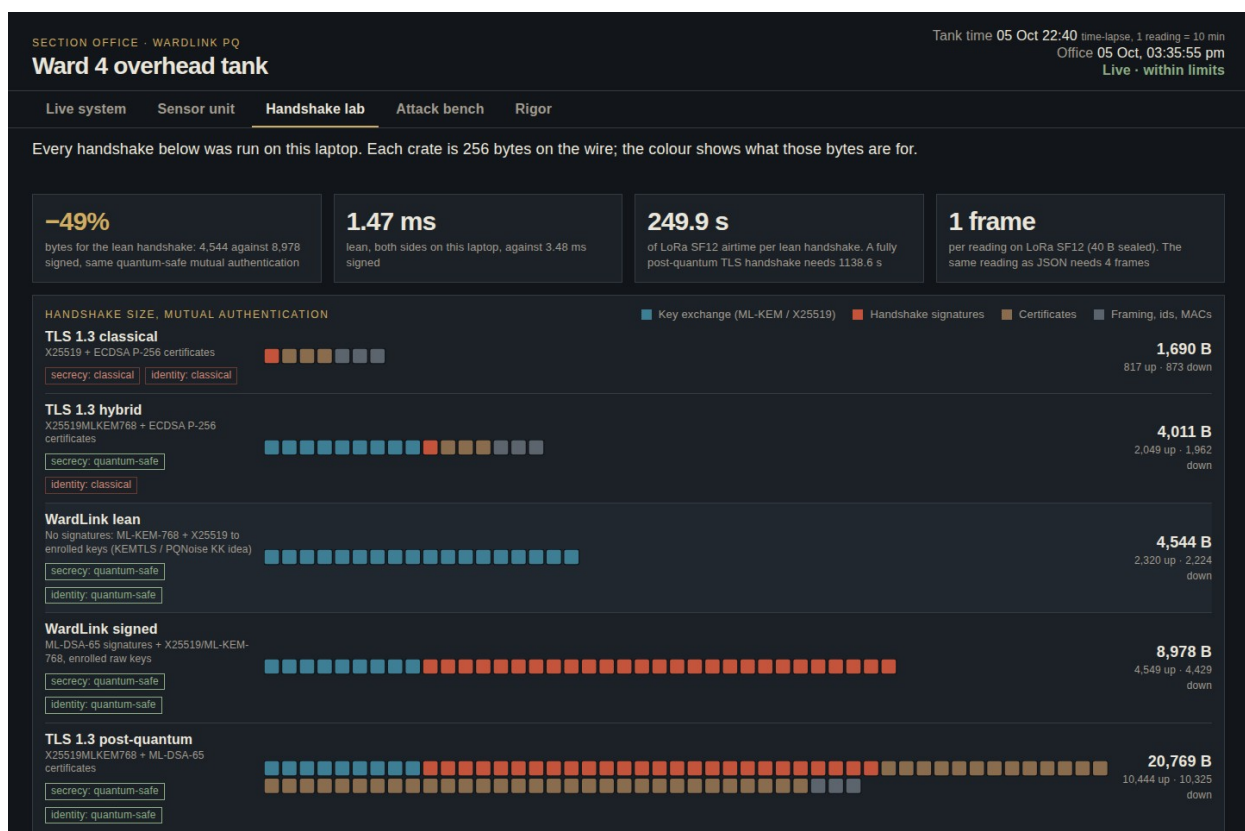


Figure 4. Handshake lab. Each crate is 256 bytes on the wire; red crates are signatures and brown ones certificates. The lean row has neither.

8.2 Field-rigor suite

The suite drives the real gateway through 40 scenarios in eight categories. Each scenario builds its own gateway on a temporary network port, so it never disturbs the live system. Latest run: 40 of 40 passed in 17.4 s.

Category	Scenarios	Examples	Result
Standards	5	NIST ACVP key-generation vectors (25 + 25), handshake round trips, fake gateway	5/5
Protocol attacks	7	Bit flip, replay, impostor, stolen board, 2,400 fuzzed messages	7/7
Transport faults	6	Stalled sender, power cut, silent connection, connection flood, 10 MB frame	6/6
Delivery	4	Outage with reconnect, lost receipt, gateway restart, full buffer	4/4
Lifecycle	6	Clone alarm, revoke, re-enrol, rate limit, watchdog, re-key without false alarm	6/6
Sensor faults	6	Frozen sensor, no echo, blind zone, dry probes, impossible pH, turbidity bubble	6/6
Physics and water	5	Plate under the sensor, rise with the main closed, IS 10500 bands, night leak, tanker forecast	5/5
Load	1	Twenty tanks reporting at once, 160 readings	1/1

SECTION OFFICE · WARDLINK PQ

Tank time 05 Oct 22:40 time-lapse, 1 reading = 10 min
Office 05 Oct, 03:35:57 pm
Live · within limits

Ward 4 overhead tank

[Live system](#)
[Sensor unit](#)
[Handshake lab](#)
[Attack bench](#)
[Rigor](#)

Forty field scenarios, from NIST conformance to power cuts, clones and twenty tanks at once. Each one builds its own gateway on a throwaway port, so running them never disturbs the live tank.

40 / 40
field scenarios passed

17.4 s
for the whole suite on this laptop

8
categories, from NIST vectors to a 20-tank load test

2026-10-05 09:55
last run (UTC) · x86_64 · Python 3.14.4

Run all scenarios now

STANDARDS · 5/5

The cryptography matches the published standard, byte for byte.

Scenario	What happened	Time
pass ML-KEM-768 key generation matches NIST ACVP vectors	25 of 25 official NIST test cases produce the expected public key	0.01 s
pass ML-DSA-65 key generation matches NIST ACVP vectors	25 of 25 official NIST test cases produce the expected public key	0.01 s
pass Lean handshake agrees on one session, 4,544 bytes	2320 + 2224 = 4544 bytes, both sides derived session eabf1142	0.01 s
pass Signed handshake agrees on one session, 8,978 bytes	4549 + 4429 = 8978 bytes with ML-DSA-65 on both sides	0.01 s
pass A fake gateway cannot prove itself to the board	Board refused the welcome: gateway could not prove its enrolled keys	0.00 s

PROTOCOL ATTACKS · 7/7

Someone on the network tries to forge, replay or read readings.

Scenario	What happened	Time
pass One flipped bit is refused; the next capsule still opens	Capsule 2 refused on its Poly1305 tag; capsule 3 opened after the ratchet skipped key 2	0.00 s
pass A recorded capsule sent again is refused	Copy refused (replayed message: that reading's key is already gone); one reading stored	0.00 s
pass Impostor board with its own keys is refused at its first capsule	Forged "tank 97%" refused; live session untouched	0.01 s
pass A board missing from the roster is refused before key work	A board calling itself tank-77 is not on the ward roster. It was refused before any key work.	0.00 s
pass A recorded hello gets the attacker nothing	5 replays answered but none could seal a capsule; 5 refused by the handshake rate limit; live session untouched	0.01 s
pass Stolen board opens no earlier reading; re-key locks it out	Opened 0 of 5 earlier capsules; read capsule [6] after the theft; re-key healed it	0.01 s
pass 2,400 mutated hellos and capsules: no crash, nothing forged	0 crashes, 0 forged capsules accepted	0.22 s

Figure 5. Rigor tab. The whole suite can be run from the dashboard; results are grouped by category with what happened in each scenario.

8.3 Problems found and fixed

- **A half-sent message froze the whole gateway.** A connection that sent part of a message and stopped blocked every tank. Fixed with one thread per connection and time limits; in the test the real sensor was served in 0.3 s while the stalled connection was dropped.
- **A power cut locked the board out.** A board that lost power right after connecting could not get back in, because the gateway kept waiting on the dead connection. Fixed the same way; the rebooted board now reconnects within 0.2 s.

9. Implementation

Part	Technology
Language	Python 3.14
Cryptography	PyCA cryptography 50 (OpenSSL-backed): ML-KEM-768, ML-DSA-65, X25519, ChaCha20-Poly1305, HKDF
Storage	SQLite in write-ahead-log mode
Gateway network	TCP with length-prefixed frames; one thread per connection
Dashboard	Standard-library web server; HTML, CSS and JavaScript with animated SVG; no external frameworks
Testing	35 unit tests and the 40-scenario field suite

Module	Responsibility
crypto.py · channel.py	Key generation, both handshakes, session keys and the per-reading key chain
record.py	The 20-byte reading and the sealed receipts
office.py · store.py	Gateway state for every tank, clone alarm, attack bench, database
rules.py · world.py	IS 10500, physics, leak, tanker and sensor-fault rules; the simulated tank
service.py · web.py	Gateway network service, sensor loop with outbox, dashboard API
bench.py · rigor.py	Handshake benchmark against TLS 1.3; the field-scenario suite

10. Hardware for the field version

Component	Purpose
ESP32-S3 board	Reads the probes, runs ML-KEM-768 and the sealing; Secure Boot and flash encryption protect its keys
Waterproof ultrasonic sensor (JSN-SR04T class)	Measures the distance to the water surface from under the roof
Air and water thermometers (DS18B20)	Correct the speed of sound and the TDS reading
Turbidity probe (optical)	Detects dirty water by scattered infrared light
TDS probe (conductivity)	Measures dissolved solids in mg/L
pH probe (glass electrode)	Measures acidity; needs periodic calibration
Analog signal isolator	Stops the TDS and pH probes disturbing each other
Li-ion cell, solar charger and 6 V panel	Power without mains on the roof
LoRa (IN865) or NB-IoT module	Long-range radio for the field
IP65 enclosure and cable glands	Weather protection
Raspberry Pi 4 (gateway)	Section-office gateway and dashboard

All components are low-cost and commonly available in India; the final selection follows the bench trials in Phase 3.

11. How we proceed

Phase	Goal	Deliverables	Duration
1. Software prototype	Prove the protocol, rules and dashboard	Gateway, tank simulator, dashboard, benchmark, 40-scenario suite	Completed
2. Hardware port	Run the sensor side on an ESP32-S3	Firmware with ML-KEM-768, outbox in flash, Secure Boot and flash encryption; measured cycles, memory and energy	6–8 weeks
3. Bench and radio trials	Validate on real links	LoRa (IN865) and NB-IoT trials, 24-hour soak test with injected loss, clock-drift check, NIST encapsulation and signing vectors, formal model of the handshake	4–6 weeks
4. Pilot	One ward tank with the municipal water department	Installed unit and office gateway, crew procedures, real data to tune thresholds, response-time measurement	8–12 weeks
5. Scale-up and handover	Many tanks, robust operation	Multi-tank roster tools, second gateway with fail-over, signed firmware updates, documentation, external security review	After the pilot

Durations are indicative and depend on hardware availability and pilot-site access.

11.1 Resources needed

- A hardware kit: sensor-unit components, a Raspberry Pi gateway and LoRa modules.
- Access to one municipal tank for the pilot, with a contact in the water department.
- A reference turbidity meter and pH buffer solutions for calibration.

11.2 Decisions requested from supervisors

- Approval to procure the hardware kit for Phase 2.
- Help identifying a pilot site and a water-department contact.
- Guidance on the DSCI submission, publication and any intellectual-property filing.

12. Limitations and risks

Limitation	Impact	Mitigation
Laptop timings and a simulated tank	Real radio loss, flash wear and battery behaviour are untested	Hardware port (Phase 2) and pilot (Phase 4)
The handshake's security argument is informal	No machine-checked proof yet	Formal model in Tamarin or ProVerif (Phase 3)
A sustained connection flood can delay a real board	The gateway is protected, but the link is not	The board retries with backoff; flood connections time out
The board buffer holds one day of readings	A longer outage drops the oldest readings	Every drop is counted and reported; a larger flash buffer in the field build
Clone detection needs both boards transmitting	A thief holding the only board leaves no overlap	Silence watchdog plus the stolen-board procedure: re-key, then re-enrol
Confirmation delays a real turbidity alarm by one reading	Ten minutes later, unless above 15 NTU	A deliberate trade-off against false alarms from bubbles
Low-cost turbidity probes cannot resolve 1 NTU precisely	Small exceedances may be missed	Lab confirmation and a reference meter for calibration
The board clock is trusted for reading times	A drifting clock could reorder readings	Clock-drift check against the gateway (Phase 3)

13. Conclusion

WardLink PQ shows that post-quantum security does not have to be heavy. By using the fact that tank sensors are enrolled in person, it halves the size of a quantum-safe handshake, seals every reading with

its own key, and delivers every reading even through outages and attacks. Combined with IS 10500 and tank-physics checks, it turns raw sensor data into trustworthy, actionable decisions for India's water supply. The software prototype is complete and verified; the next step is the hardware port and a field pilot.

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Appendix A. Running the prototype and a five-minute demonstration

From the project folder, with the Python environment active:

Command	What it does
<code>python -m wardlink demo</code>	Starts the gateway, the simulated tank sensor and the dashboard at http://127.0.0.1:8765
<code>python -m wardlink rigor</code>	Runs the 40 field scenarios and saves data/rigor.json
<code>python -m wardlink bench</code>	Measures the five handshakes and their radio cost
<code>python -m wardlink check</code>	Quick handshake and refusal checks without the network

A suggested five-minute demonstration for reviewers:

1. **Live system:** click stages 1 to 5 to walk one reading from the ultrasonic ping to the action flag.
2. **Sensor unit:** click the ultrasonic sensor and the ESP32-S3; let the sampling cycle play.
3. **Handshake lab:** point to the red signature crates, which disappear in the lean row.
4. **Attack bench:** block the sensor with a plate (caught by physics), cut the radio link (nothing lost), clone the board (clone alarm).
5. **Rigor:** run all scenarios and show the result.